FORENSIC SKETCH RECOGNITION SYSTEM

Name of Students:
DEVANSHI T S, DHANISA R, NEIL JOE AUGUSTINE, RAHUL DAS
Project Guide: Ms. MONISHA THOMAS

Academic Year
2025-2026

Introduction

A web-based application designed to assist law enforcement agencies in the critical task of identifying suspects. The system accurately matches hand-drawn forensic sketches or photos against a database of faces, even when the input image contains significant variations in pose.

Objectives

- Develop an interactive sketch generation system using React + Fabric.js
- Extract facial features using HOG descriptors and landmark detection
- Design a hybrid matching algorithm (appearance + geometry)
- Retrieve Top-5 suspect matches with confidence scores
- Build a scalable AI-assisted forensic investigation system

Methodology

- Sketch Generation:

The system provides an interactive canvas using Fabric.js, where users can create a suspect's face by dragging and arranging facial components.

- Image Preprocessing:

The input sketch (generated or uploaded) is prepared for analysis by converting it to grayscale, resizing, and normalizing to ensure consistency with database images.

- Landmark Detection:

Using dlib, the system detects key facial landmarks (eyes, nose, mouth, jawline). These landmarks help in understanding the facial structure and isolating important regions.

- Hybrid Feature Extraction:

The system extracts HOG features and Geometric features (distances, ratios, angles) for structural representation. This combination ensures robustness against sketch variations and pose changes.

- Matching and Ranking:

The generated feature vector is compared with a database of criminal images using a hybrid similarity score (appearance + geometry). The results are ranked based on confidence scores.

- Result Visualization:

The system displays the Top-5 matching suspects in a structured interface, along with similarity scores and relevant criminal details.

System Architecture Diagram

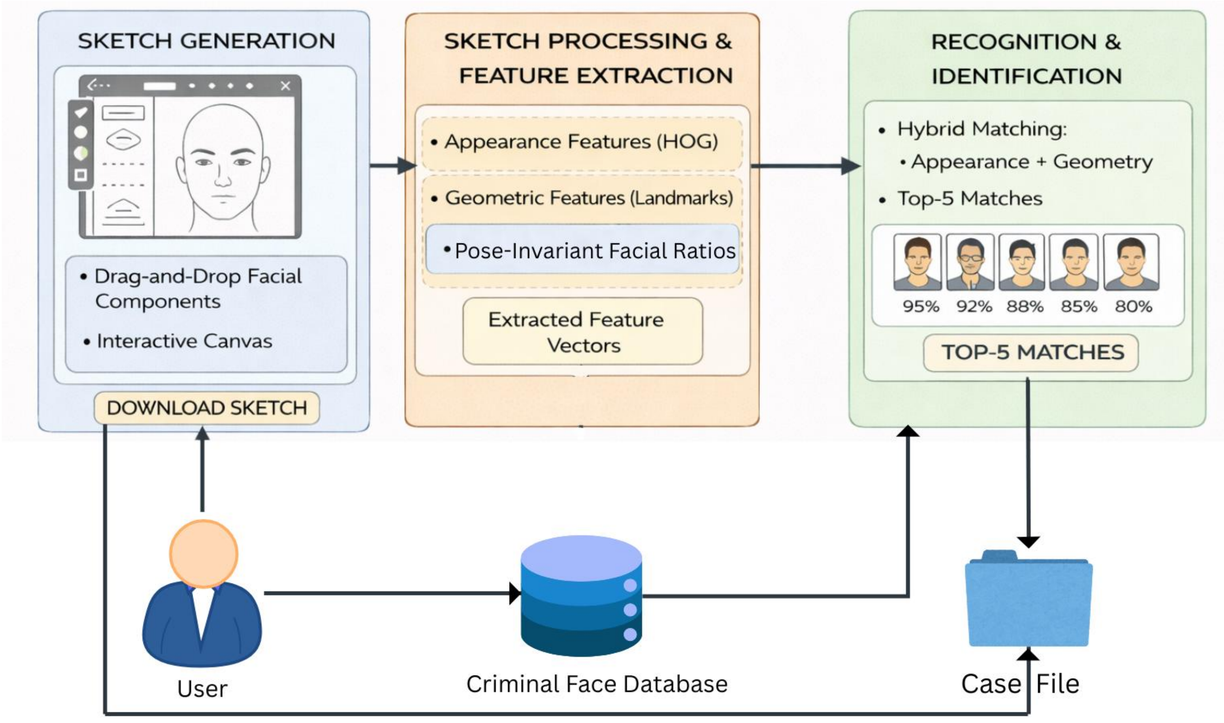

Figure. system architecture.

Block Diagram

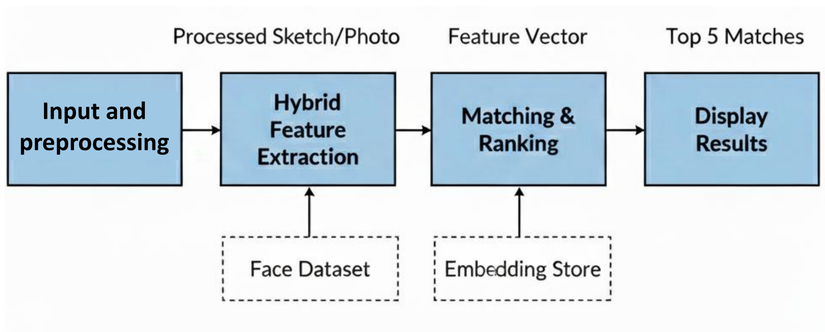

Figure. Block Diagram.

The system uses a multi-layer architecture (Frontend → API → AI Engine → Database) for efficient processing.

Results

- A ranked list of the Top 5 matching suspects is retrieved from the criminal image database, providing investigators with a concise set of high-probability candidates.
- Each matched image is displayed along with its confidence score, indicating the similarity between the input sketch and database images, helping prioritize further investigation.
- The results are presented in a clean and interactive interface, allowing users to easily view suspect details such as ID, crime record, and department without requiring technical knowledge.
- The system successfully identifies matches even for sketches with variations in proportions and missing details, demonstrating the effectiveness of the hybrid feature extraction approach.

Achievements

- Completed a comprehensive system design, including Software Requirements Specification (SRS), architecture diagrams, and modular workflow integration.
- Developed a robust pose-invariant recognition model using facial landmarks and geometric ratios to handle variations in facial structure and orientation.
- Implemented an effective hybrid feature extraction approach (HOG + geometric features) to bridge the gap between sketches and real images.
- Designed and built a user-friendly interactive interface for sketch generation and identification, ensuring ease of use for investigators with minimal technical expertise.

Conclusion

This Forensic Sketch Recognition System provides a practical and effective solution for law enforcement by integrating an interactive sketch generation module with an AI-based identification system. By utilizing a hybrid feature extraction approach combining HOG and geometric features, the system improves matching accuracy between sketches and real images.

The ability to handle variations in facial proportions and incomplete sketch details, along with a user-friendly web interface, makes it a valuable tool for accelerating suspect identification and supporting modern forensic investigations.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
(NBA Accredited)

© Poster Template by Genigraphics® 1.800.790.4001 www.genigraphics.com